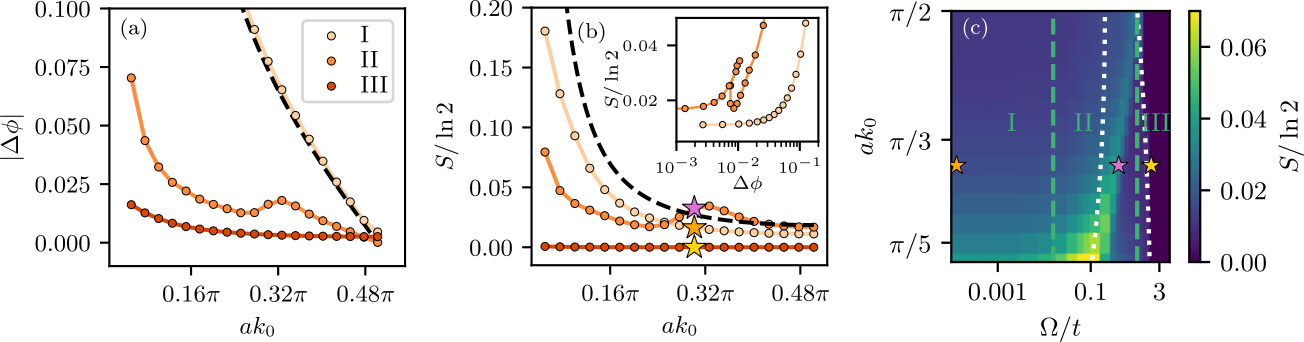

(a)
I
II
III
$|\Delta\phi|$
0.100
0.075
0.050
0.025
0.000
0.16π
0.32π
0.48π
ak_0
(b)
$S/\ln 2$
0.20
0.15
0.10
0.05
0.00
0.04
0.02
10^{-3}
10^{-2}
10^{-1}
$\Delta\phi$
(c)
ak_0
π/2
π/3
π/5
0.001
0.1
3
Ω/t
0.06
0.04
0.02
0.00